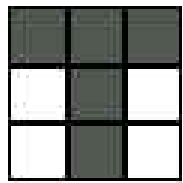


NN on Pattern Recognition

- ❑ An important application of neural networks is pattern recognition. Pattern recognition can be implemented by using a feed-forward neural network that has been trained accordingly.
- ❑ During training, the network is trained to associate outputs with input patterns.
- ❑ When the network is used, it identifies the input pattern and tries to output the associated output pattern.
- ❑ When an input pattern has no output associated with it, the network gives the output that corresponds to a taught input pattern that is least different from the given pattern.

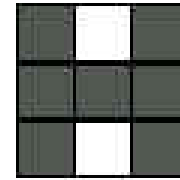
Pattern Recognition Example



INPUT



OUTPUT



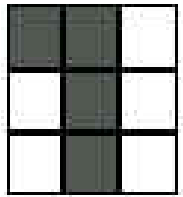
INPUT



OUTPUT

- Suppose a network is trained to recognize the patterns T and H. The associated patterns are all black and all white respectively as shown above.

Pattern Recognition Example (Cont.)

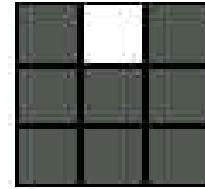


INPUT

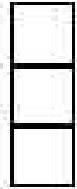


OUTPUT

Since the input pattern looks more like a 'T', when the network classifies it, it sees the input closely resembling 'T' and outputs the pattern that represents a 'T'.



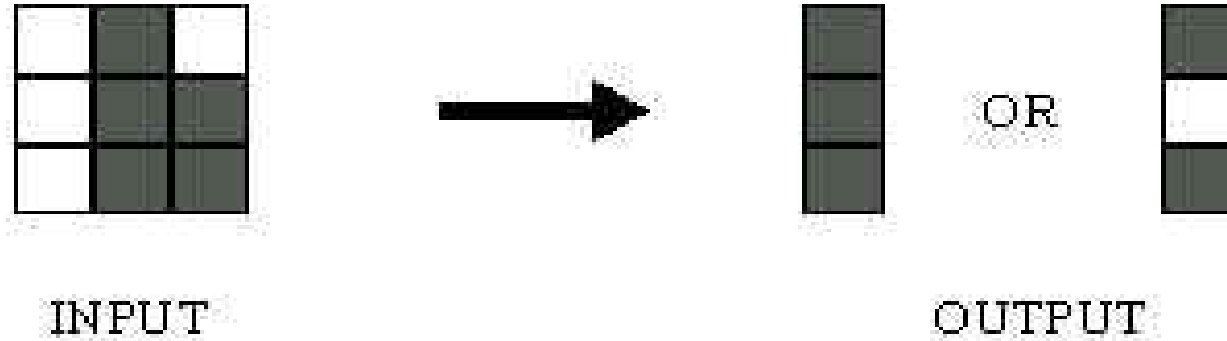
INPUT



OUTPUT

The input pattern here closely resembles 'H' with a slight difference. The network in this case classifies it as an 'H' and outputs the pattern representing an 'H'.

Pattern Recognition Example (Cont.)



- ❑ Here the top row is 2 errors away from a 'T' and 3 errors away from an H. So the top output is a black.
- ❑ The middle row is 1 error away from both T and H, so the output is random.
- ❑ The bottom row is 1 error away from T and 2 away from H. Therefore the output is black.
- ❑ Since the input resembles a 'T' more than an 'H' the output of the network is in favor of a 'T'.